CAS SciFinder

Task History

Initiating Search

November 10, 2025, 3:54 PM

Search:

CAS Lexicon Search:

N-Chlorosuccinimide - Preferred Concept

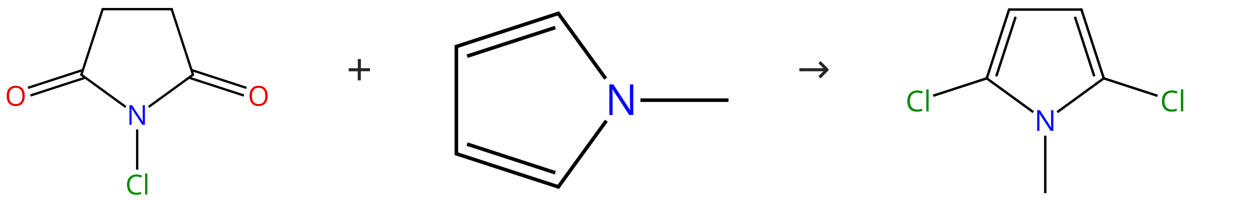
Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (4,885)	**References**	View Results
Exported: Retrieved Related Reaction Results + Filters (2,534) Filtered By: Yield: **90-100%, 80-89%, 70-79%** Document Type: **Journal**	**Reactions**	View Results

Reactions (50)

View in CAS SciFinder

Scheme 1 (1 Reaction) — Steps: **1** Yield: **100%**



Suppliers (99) | Suppliers (75) | Suppliers (5)

31-084-CAS-8970901 — Steps: **1** Yield: **100%**

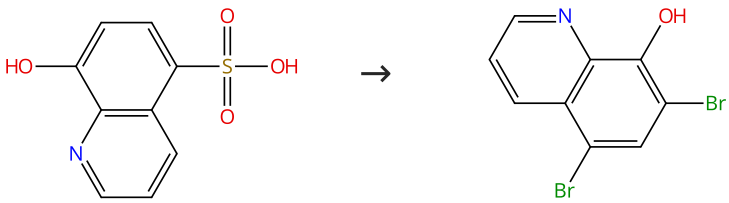
No Data Available

Bromination and chlorination of pyrrole and some reactive 1-substituted pyrroles

By: Gilow, H. M.; et al

Journal of Organic Chemistry (1981), 46(11), 2221-5.

Scheme 2 (1 Reaction) — Steps: **1** Yield: **100%**



Suppliers (59) | Suppliers (81)

31-084-CAS-3949451 — Steps: **1** Yield: **100%**

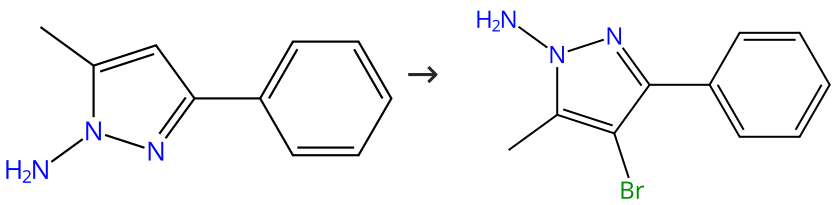
1.1 **Reagents:** *N*-Bromosuccinimide
Solvents: Acetic acid, Acetonitrile; 10 min, 60 °C

Effect of N-halosucinimides on 5-, 7- and 5,7-8-quinolinol sulfonic acids

By: Gershon, Herman; et al

Trends in Heterocyclic Chemistry (2007), 12, 67-73.

Scheme 3 (1 Reaction) — Steps: **1** Yield: **100%**



Suppliers (4) | Supplier (1)

31-084-CAS-5539733 — Steps: **1** Yield: **100%**

1.1 **Reagents:** Bromine
Solvents: Dichloromethane

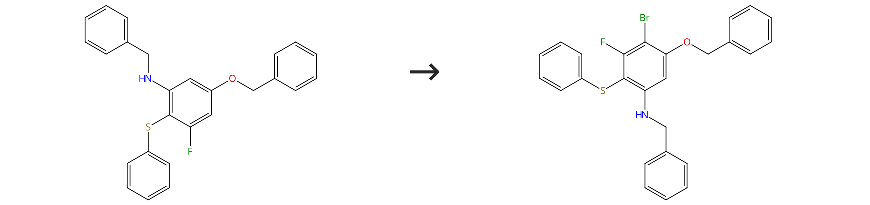
Reactions of N-aminopyrazoles with halogenating reagents and synthesis of 1,2,3-triazines

By: Osawa, Akio; et al

Chemical & Pharmaceutical Bulletin (1988), 36(10), 3838-48.

Scheme 4 (1 Reaction)

Steps: **1** Yield: **100%**



31-084-CAS-13866343

Steps: **1** Yield: **100%**

1.1 **Reagents:** *N*-Bromosuccinimide
Solvents: Dichloromethane; 2.5 h, 0 °C

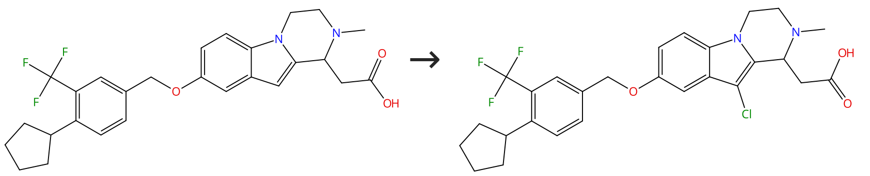
Regioselective approach to multi-substituted benzenes

By: Seo, Hana; et al

Chemistry Letters (2011), 40(7), 744-746.

Scheme 5 (1 Reaction)

Steps: **1** Yield: **100%**



Supplier (1)

31-084-CAS-2821953

Steps: **1** Yield: **100%**

1.1 **Reagents:** Chlorosuccinimide
Solvents: Dichloromethane; 0 °C → rt

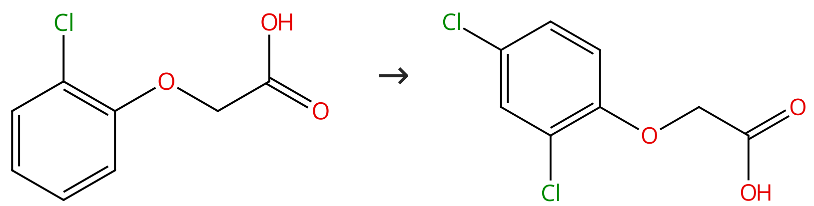
Design and synthesis of new tricyclic indoles as potent modulators of the $S1P_1$ receptor

By: Buzard, Daniel J.; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(3), 659-663.

Scheme 6 (1 Reaction)

Steps: **1** Yield: **100%**



Suppliers (86)

Suppliers (100)

31-084-CAS-23031761

Steps: **1** Yield: **100%**

1.1 **Reagents:** *N*-Chloro-*N*-(phenylsulfonyl)benzenesulfonamide
Solvents: Acetonitrile; 5 min, 20 - 25 °C

1.2 **Reagents:** Water
Solvents: Dichloromethane; 10 - 15 min, 20 - 25 °C

Experimental Protocols

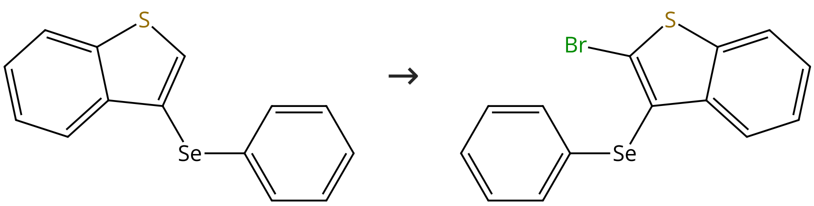
Activator free, expeditious and eco-friendly chlorination of activated arenes by N-chloro-N-(phenylsulfonyl)benzene sulfonamide (NCBSI)

By: Misal, Balu; et al

Tetrahedron Letters (2021), 63, 152689.

Scheme 7 (1 Reaction)

Steps: **1** Yield: **100%**



31-084-CAS-15121543 Steps: **1** Yield: **100%**

1.1 **Reagents:** Butyllithium
Solvents: Tetrahydrofuran, Hexane; 0 °C; 0 °C → rt; 1 h, rt

1.2 **Reagents:** 1,2-Dibromo-1,1,2,2-tetrachloroethane; 0 °C; 6 h, rt

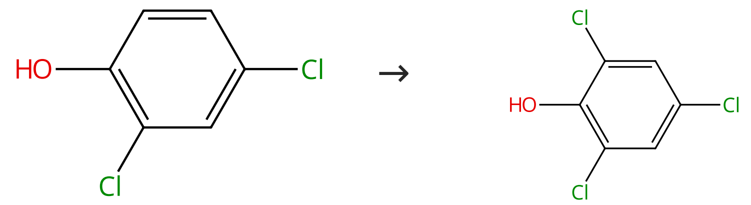
Experimental Protocols

Consecutive Thiophene-Annulation Approach to π-Extended Thienoacene-Based Organic Semiconductors with [1]Benzothieno[3,2-b][1]benzothiophene (BTBT) Substructure

By: Mori, Takamichi; et al

Journal of the American Chemical Society (2013), 135(37), 13900-13913.

Scheme 8 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (99) Suppliers (87)

31-084-CAS-23033350 Steps: **1** Yield: **100%**

1.1 **Reagents:** *N*-Chloro-*N*-(phenylsulfonyl)benzenesulfonamide
Solvents: Acetonitrile; 10 min, 20 - 25 °C

1.2 **Reagents:** Water
Solvents: Dichloromethane; 10 - 15 min, 20 - 25 °C

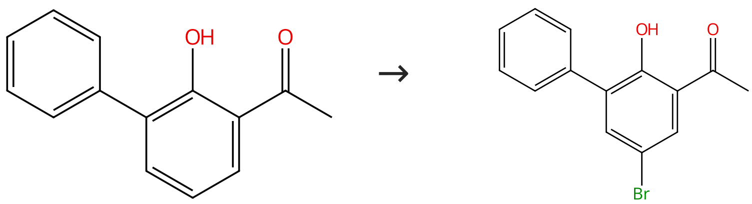
Experimental Protocols

Activator free, expeditious and eco-friendly chlorination of activated arenes by N-chloro-N-(phenylsulfonyl)benzene sulfonamide (NCBSI)

By: Misal, Balu; et al

Tetrahedron Letters (2021), 63, 152689.

Scheme 9 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (44) Suppliers (2)

31-084-CAS-9837269 Steps: **1** Yield: **100%**

1.1 **Reagents:** *N*-Bromosuccinimide
Solvents: Acetic acid; 2 h, 85 °C

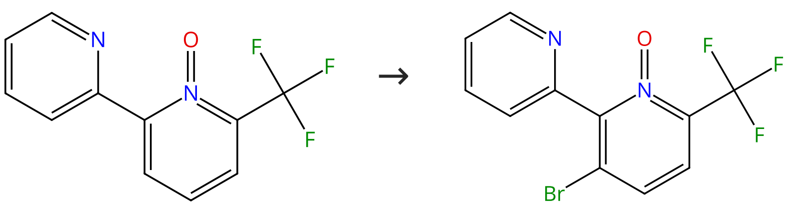
Experimental Protocols

Development of serine protease inhibitors displaying a multicentered short (<2.3 Å) hydrogen bond binding mode: Inhibitors of urokinase-type plasminogen activator and factor Xa

By: Verner, Erik; et al

Journal of Medicinal Chemistry (2001), 44(17), 2753-2771.

Scheme 10 (1 Reaction) Steps: **1** Yield: **100%**



Supplier (1)

31-084-CAS-17240926 Steps: **1** Yield: **100%**

1.1 **Reagents:** *N*-Bromosuccinimide
Catalysts: Palladium diacetate
Solvents: Chlorobenzene; 24 h, 110 °C

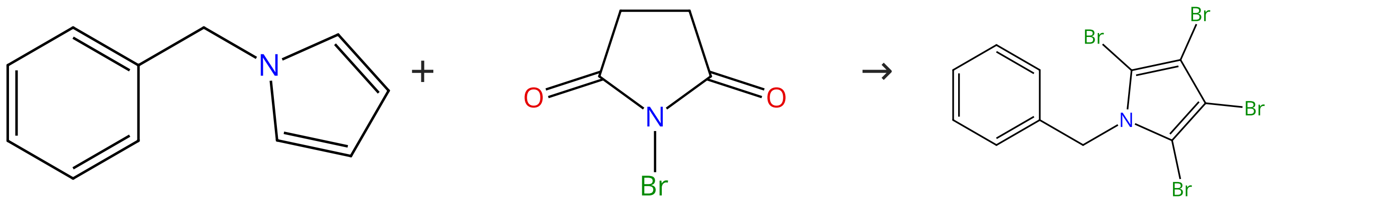
Experimental Protocols

Palladium-Catalyzed Directed Halogenation of Bipyridine N-Oxides

By: Zucker, Sina P.; et al

Journal of Organic Chemistry (2017), 82(11), 5616-5635.

Scheme 11 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (69) Suppliers (101) Supplier (1)

31-084-CAS-10449571 Steps: **1** Yield: **100%**

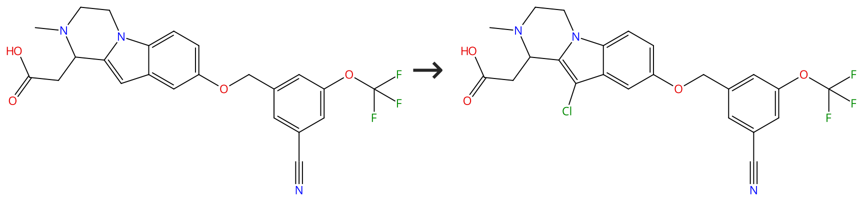
No Data Available

Bromination and chlorination of pyrrole and some reactive 1-substituted pyrroles

By: Gilow, H. M.; et al

Journal of Organic Chemistry (1981), 46(11), 2221-5.

Scheme 12 (1 Reaction) Steps: **1** Yield: **100%**



Supplier (1)

31-084-CAS-8935179 Steps: **1** Yield: **100%**

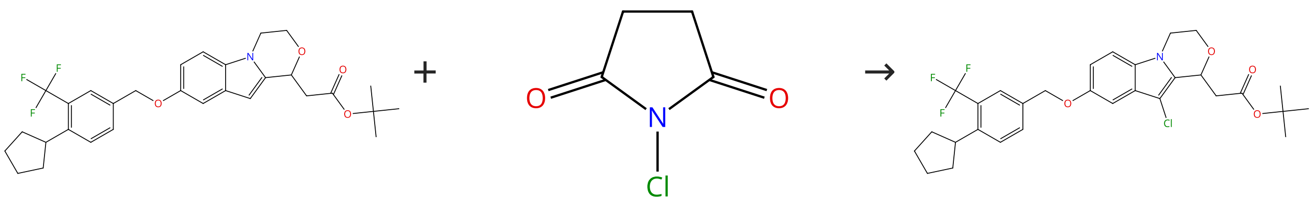
1.1 **Reagents:** Chlorosuccinimide
Solvents: Dichloromethane; 0 °C → rt

Design and synthesis of new tricyclic indoles as potent modulators of the $S1P_1$ receptor

By: Buzard, Daniel J.; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(3), 659-663.

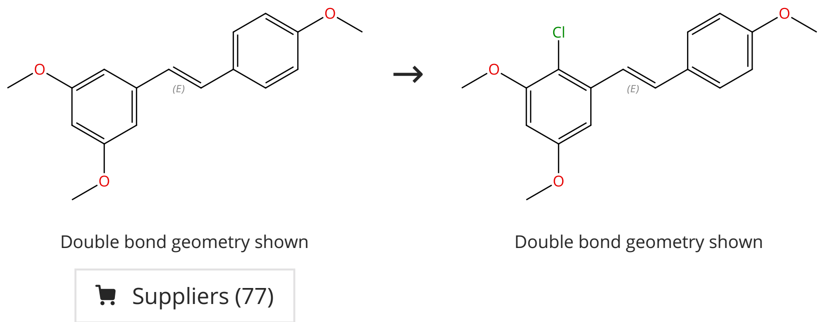
Scheme 13 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (99)

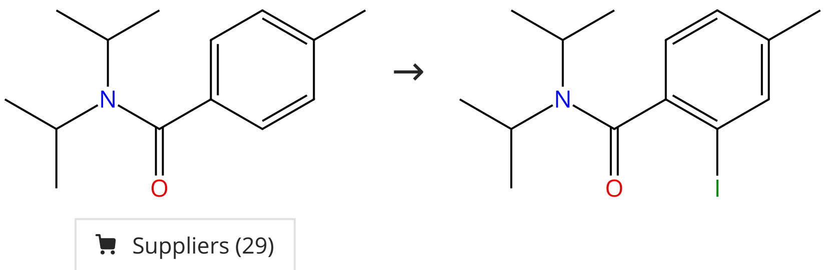
31-084-CAS-18183871	Steps: **1** Yield: **100%**	**Design and synthesis of new tricyclic indoles as potent modulators of the S1P$_1$ receptor**
1.1	**Solvents:** Dichloromethane; 0 °C → rt	By: Buzard, Daniel J.; et al Bioorganic & Medicinal Chemistry Letters (2015), 25(3), 659-663.

Scheme 14 (1 Reaction) Steps: **1** Yield: **100%**



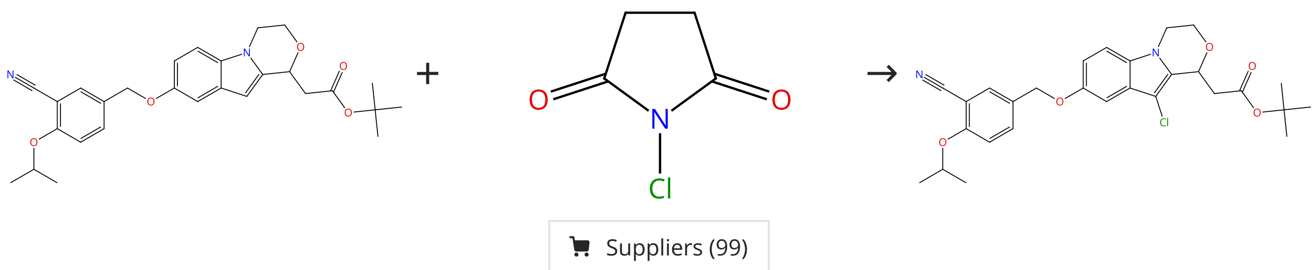
31-084-CAS-19489252	Steps: **1** Yield: **100%**	**Atom-Economic Electron Donors for Photobiocatalytic Halogenations**
1.1	**Reagents:** Potassium bromide, Vanadate (VO_4^{3-}), sodium (1:3), (*T*-4)- **Catalysts:** Flavin mononucleotide **Solvents:** Acetonitrile, Water; 16 h, pH 6, 25 °C	By: Seel, Catharina Julia; et al ChemCatChem (2018), 10(18), 3960-3963.
Experimental Protocols		

Scheme 15 (1 Reaction) Steps: **1** Yield: **100%**



31-084-CAS-22871968	Steps: **1** Yield: **100%**	**Identification of key functionalization species in the Cp*Ir(III)-catalyzed-ortho halogenation of benzamides**
1.1	**Reagents:** Acetic acid, *N*-Iodosuccinimide **Catalysts:** Silver triflate, Iridium, di-μ-chlorodichlorobis[(1,2,3,4,5-η)-1,2,3,4,5-pentamethyl-2,4-cyclopentadien-1-yl]di- **Solvents:** 1,2-Dichloroethane; 1 h, 60 °C	By: Guzman Santiago, Alexis J.; et al Dalton Transactions (2020), 49(45), 16166-16174.
Experimental Protocols		

Scheme 16 (1 Reaction) Steps: **1** Yield: **100%**



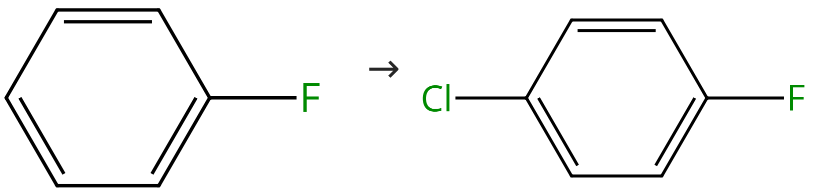
31-084-CAS-18177038	Steps: **1** Yield: **100%**
1.1	**Solvents:** Dichloromethane; 0 °C → rt

Design and synthesis of new tricyclic indoles as potent modulators of the S1P$_1$ receptor

By: Buzard, Daniel J.; et al

Bioorganic & Medicinal Chemistry Letters (2015), 25(3), 659-663.

Scheme 17 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (88)

Suppliers (60)

31-084-CAS-8621113	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium chloride, Potassium persulfate **Solvents:** Acetonitrile, Water; 1.5 h, 100 °C
1.2	**Reagents:** Sodium thiosulfate **Solvents:** Water

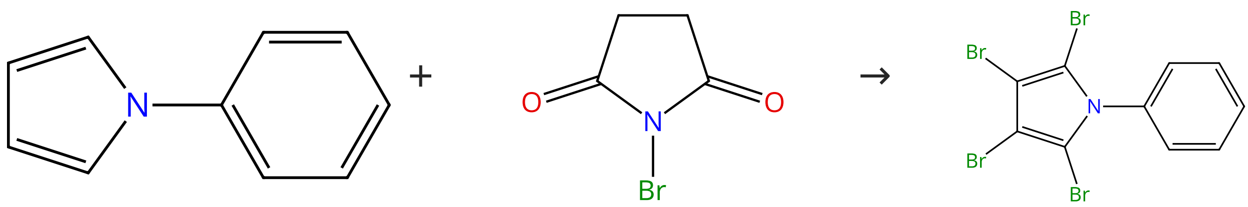
Experimental Protocols

Efficient Oxidative Chlorination of Aromatics on Saturated Sodium Chloride Solution

By: Gu, Liuqun; et al

Advanced Synthesis & Catalysis (2013), 355(6), 1077-1082.

Scheme 18 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (71)

Suppliers (101)

Suppliers (2)

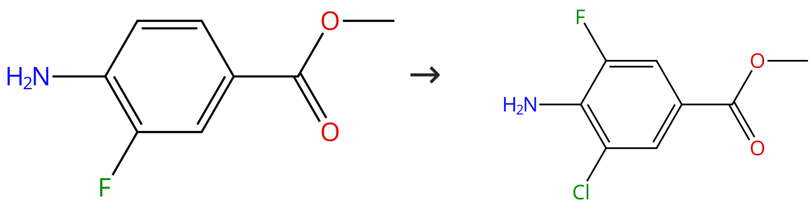
31-084-CAS-7983078	Steps: **1** Yield: **100%**
No Data Available	

Bromination and chlorination of pyrrole and some reactive 1-substituted pyrroles

By: Gilow, H. M.; et al

Journal of Organic Chemistry (1981), 46(11), 2221-5.

Scheme 19 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (62)

Suppliers (41)

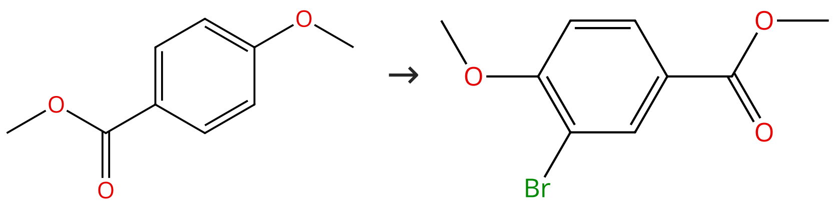
31-614-CAS-39834437	Steps: **1** Yield: **100%**
1.1	**Reagents:** Chlorosuccinimide **Solvents:** Dimethylformamide; 3 h, rt
1.2	**Reagents:** Water; 2 min, 0 °C
Experimental Protocols	

Exploring boron applications in modern agriculture: A structure-activity relationship study of a novel series of multi-substitution benzoxaboroles for identification of potential fungicides

By: Liu, Chunliang; et al

Bioorganic & Medicinal Chemistry Letters (2021), 43, 128089.

Scheme 20 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (91) Suppliers (78)

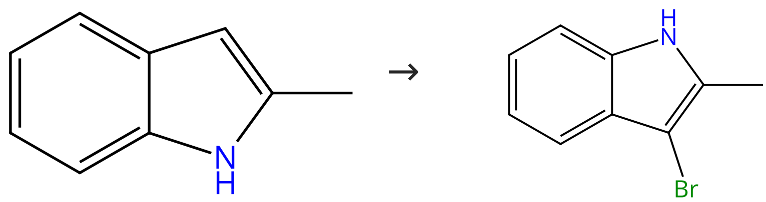
31-084-CAS-20103702	Steps: **1** Yield: **100%**
1.1	**Reagents:** *N*-Bromosuccinimide **Catalysts:** Diphenyl disulfide **Solvents:** Acetonitrile; 12 h, rt
1.2	**Reagents:** Sodium bicarbonate, Dipotassium phosphate **Solvents:** Water; rt

Aromatic Halogenation Using N-Halosuccinimide and PhSSiMe$_3$ or PhSSPh

By: Hirose, Yuuka; et al

Journal of Organic Chemistry (2019), 84(11), 7405-7410.

Scheme 21 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (90) Suppliers (39)

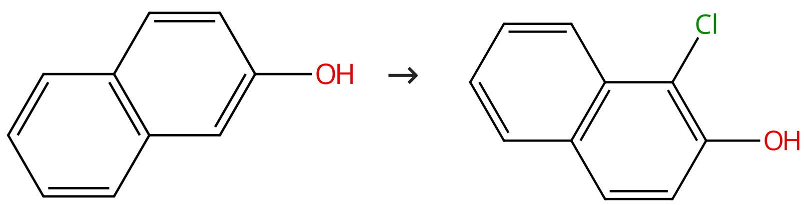
31-084-CAS-19489247	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium bromide, Vanadate (VO_4^{3-}), sodium (1:3), (*T*-4)- **Catalysts:** Flavin mononucleotide, Haloperoxidase **Solvents:** Acetonitrile, Water; 16 h, pH 6, 25 °C
Experimental Protocols	

Atom-Economic Electron Donors for Photobiocatalytic Halogenations

By: Seel, Catharina Julia; et al

ChemCatChem (2018), 10(18), 3960-3963.

Scheme 22 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (117) Suppliers (60)

31-084-CAS-23032184 Steps: **1** Yield: **100%**

1.1 **Reagents:** *N*-Chloro-*N*-(phenylsulfonyl)benzenesulfonamide
Solvents: Acetonitrile; 20 min, 20 - 25 °C

1.2 **Reagents:** Water
Solvents: Dichloromethane; 10 - 15 min, 20 - 25 °C

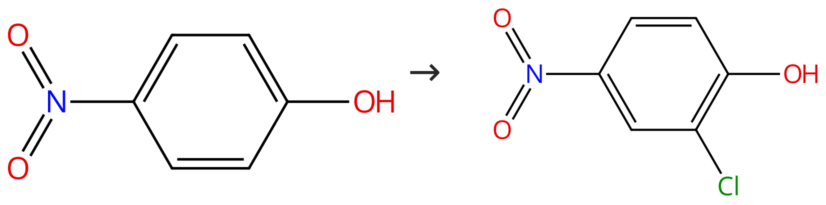
Experimental Protocols

Activator free, expeditious and eco-friendly chlorination of activated arenes by N-chloro-N-(phenylsulfonyl)benzene sulfonamide (NCBSI)

By: Misal, Balu; et al

Tetrahedron Letters (2021), 63, 152689.

Scheme 23 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (133) Suppliers (86)

31-084-CAS-23029533 Steps: **1** Yield: **100%**

1.1 **Reagents:** *N*-Chloro-*N*-(phenylsulfonyl)benzenesulfonamide
Solvents: Acetonitrile; 10 min, 20 - 25 °C

1.2 **Reagents:** Water
Solvents: Dichloromethane; 10 - 15 min, 20 - 25 °C

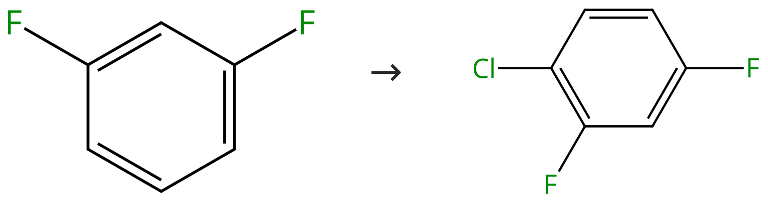
Experimental Protocols

Activator free, expeditious and eco-friendly chlorination of activated arenes by N-chloro-N-(phenylsulfonyl)benzene sulfonamide (NCBSI)

By: Misal, Balu; et al

Tetrahedron Letters (2021), 63, 152689.

Scheme 24 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (61) Suppliers (76)

31-084-CAS-6497966 Steps: **1** Yield: **100%**

1.1 **Reagents:** Sodium chloride, Potassium persulfate
Solvents: Acetonitrile, Water; 1.5 h, 100 °C

1.2 **Reagents:** Sodium thiosulfate
Solvents: Water

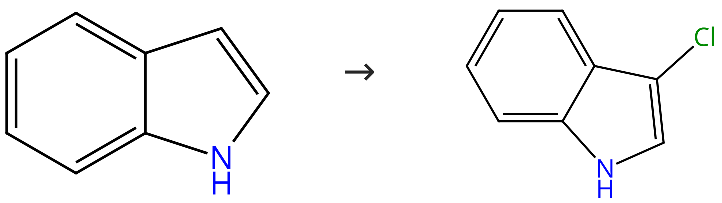
Experimental Protocols

Efficient Oxidative Chlorination of Aromatics on Saturated Sodium Chloride Solution

By: Gu, Liuqun; et al

Advanced Synthesis & Catalysis (2013), 355(6), 1077-1082.

Scheme 25 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (113) Suppliers (48)

31-084-CAS-6770321 Steps: **1** Yield: **100%**

1.1 **Reagents:** 1,4-Dimethylpiperazine, Chlorosuccinimide
Solvents: Dichloromethane; 2 h, 0 °C

1.2 **Reagents:** Trichloroacetic acid, 3-Methyl-2-buten-1-ol
Solvents: Dichloromethane; 1 h, 0 °C; overnight, rt

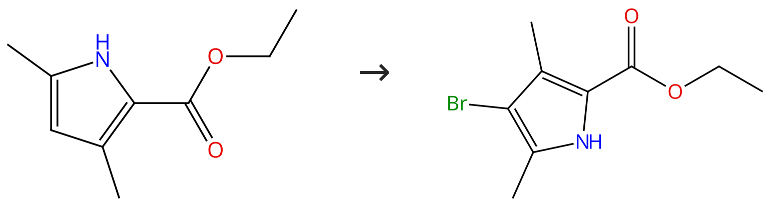
Experimental Protocols

Step-economic synthesis of (±)-debromoflustramine A using indole C3 activation strategy

By: Ignatenko, Vasily A.; et al

Tetrahedron Letters (2011), 52(12), 1269-1272.

Scheme 26 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (85) Suppliers (54)

31-084-CAS-19489243 Steps: **1** Yield: **100%**

1.1 **Reagents:** Potassium bromide, Vanadate (VO_4^{3-}), sodium (1:3), (*T*-4)-
Catalysts: Flavin mononucleotide, Haloperoxidase
Solvents: Acetonitrile, Water; 16 h, pH 6, 25 °C

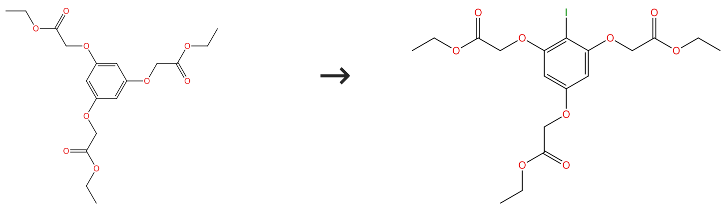
Experimental Protocols

Atom-Economic Electron Donors for Photobiocatalytic Halogenations

By: Seel, Catharina Julia; et al

ChemCatChem (2018), 10(18), 3960-3963.

Scheme 27 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (2)

31-614-CAS-24202727 Steps: **1** Yield: **100%**

1.1 **Reagents:** Chlorosuccinimide, Sodium iodide
Solvents: Acetic acid; 1.5 h, rt

1.2 **Reagents:** Sodium bicarbonate
Solvents: Water; neutralized, rt

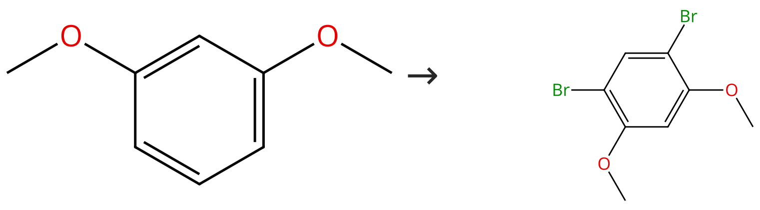
Experimental Protocols

Design of novel, water soluble and highly luminescent europium labels with potential to enhance immunoassay sensitivities

By: Sund, Henri; et al

Molecules (2017), 22(10), 1807/1-1807/23.

Scheme 28 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (88) Suppliers (58)

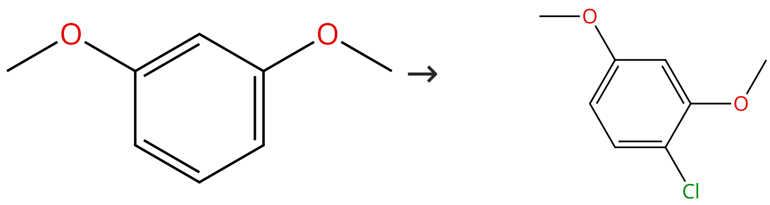
31-084-CAS-20103708	Steps: **1** Yield: **100%**
1.1	**Reagents:** *N*-Bromosuccinimide **Catalysts:** Diphenyl disulfide **Solvents:** Acetonitrile; 1 h, rt
1.2	**Reagents:** Sodium bicarbonate, Dipotassium phosphate **Solvents:** Water; rt

Aromatic Halogenation Using N-Halosuccinimide and PhSSiMe$_3$ or PhSSPh

By: Hirose, Yuuka; et al

Journal of Organic Chemistry (2019), 84(11), 7405-7410.

Scheme 29 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (88) Suppliers (57)

31-084-CAS-19489250	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium bromide, Vanadate (VO_4^{3-}), sodium (1:3), (*T*-4)- **Catalysts:** Flavin mononucleotide **Solvents:** Acetonitrile, Water; 16 h, pH 6, 25 °C

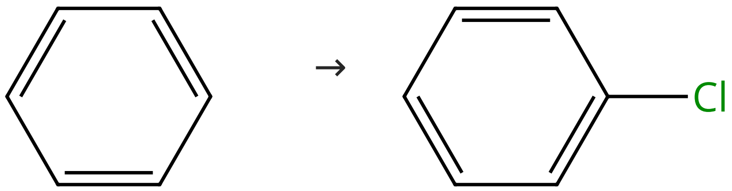
Experimental Protocols

Atom-Economic Electron Donors for Photobiocatalytic Halogenations

By: Seel, Catharina Julia; et al

ChemCatChem (2018), 10(18), 3960-3963.

Scheme 30 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (174) Suppliers (127)

31-084-CAS-15017849	Steps: **1** Yield: **100%**
1.1	**Reagents:** Sodium chloride, Potassium persulfate **Solvents:** Acetonitrile, Water; 1 h, 100 °C
1.2	**Reagents:** Sodium thiosulfate **Solvents:** Water

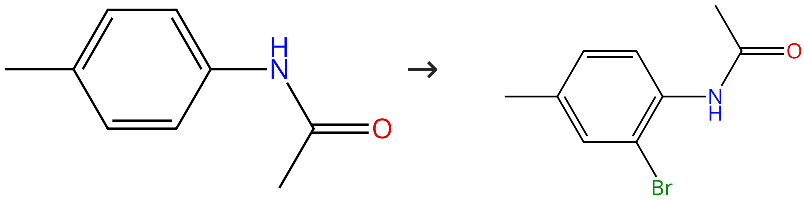
Experimental Protocols

Efficient Oxidative Chlorination of Aromatics on Saturated Sodium Chloride Solution

By: Gu, Liuqun; et al

Advanced Synthesis & Catalysis (2013), 355(6), 1077-1082.

Scheme 31 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (75) Suppliers (68)

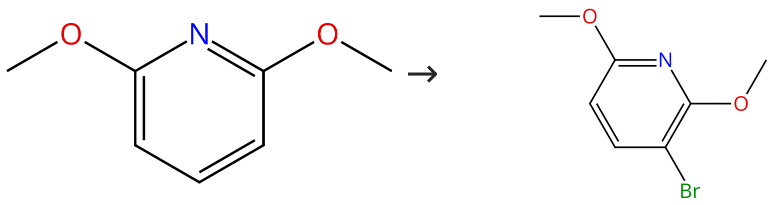
31-084-CAS-20103710	Steps: **1** Yield: **100%**
1.1	**Reagents:** *N*-Bromosuccinimide **Catalysts:** Diphenyl disulfide **Solvents:** Acetonitrile; 1 h, rt
1.2	**Reagents:** Sodium bicarbonate, Dipotassium phosphate **Solvents:** Water; rt

Aromatic Halogenation Using N-Halosuccinimide and PhSSiMe$_3$ or PhSSPh

By: Hirose, Yuuka; et al

Journal of Organic Chemistry (2019), 84(11), 7405-7410.

Scheme 32 (1 Reaction) Steps: **1** Yield: **100%**



Suppliers (74) Suppliers (65)

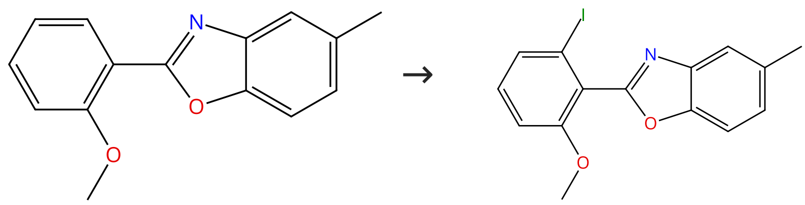
31-084-CAS-19489241	Steps: **1** Yield: **100%**
1.1	**Reagents:** Potassium bromide, Vanadate (VO_4^{3-}), sodium (1:3), (*T*-4)- **Catalysts:** Flavin mononucleotide, Haloperoxidase **Solvents:** Acetonitrile, Water; 6 h, pH 6, 25 °C
Experimental Protocols	

Atom-Economic Electron Donors for Photobiocatalytic Halogenations

By: Seel, Catharina Julia; et al

ChemCatChem (2018), 10(18), 3960-3963.

Scheme 33 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (22)

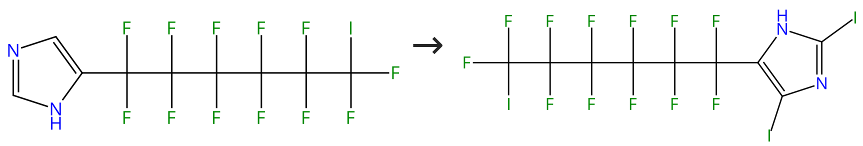
31-084-CAS-20456593	Steps: **1** Yield: **99%**
1.1	**Reagents:** Acetic acid, *N*-Iodosuccinimide **Catalysts:** Rhodium(2+), tris(acetonitrile)[(1,2,3,4,5-η)-1,2,3,4,5-pentamethyl-2,4-cyclopentadien-1-yl]-, (*OC*-6-11)-hexafluoroantimonate(1-) (1:2) **Solvents:** 1,2-Dichloroethane; -196 °C; -196 °C → rt; 18 h, 60 °C

Transition metal catalyzed C7 and ortho- selective halogenation of 2-arylbenzo[d]oxazoles

By: Hong, Xi; et al

Organic Chemistry Frontiers (2019), 6(13), 2226-2233.

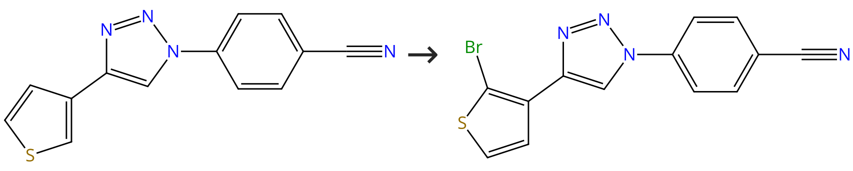
Scheme 34 (1 Reaction) Steps: **1** Yield: **99%**



Supplier (1) Supplier (1)

31-084-CAS-8441062	Steps: **1** Yield: **99%**	**Synthesis of (ω-haloperfluoroalkyl)imidazoles and their derivatives**
1.1 **Reagents:** Iodine		By: Fujii, Shozo; et al Nagoya Kogyo Gijutsu Shikensho Hokoku (1986), 35(3), 117-29.

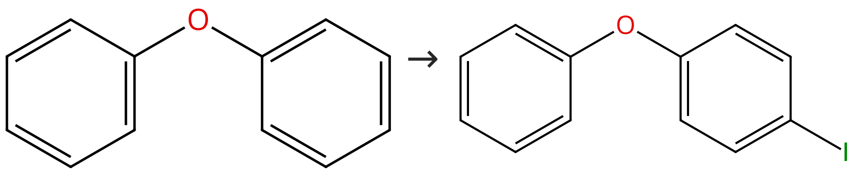
Scheme 35 (1 Reaction) Steps: **1** Yield: **99%**



Supplier (1)

31-084-CAS-16741027	Steps: **1** Yield: **99%**	**Selective C(sp^2)-H Halogenation of "Click" 4-Aryl-1,2,3-triazoles**
1.1 **Reagents:** *N*-Bromosuccinimide **Solvents:** 1,2-Dichloroethane; 18 h, rt		By: Goitia, Asier; et al Organic Letters (2017), 19(4), 962-965.
Experimental Protocols		

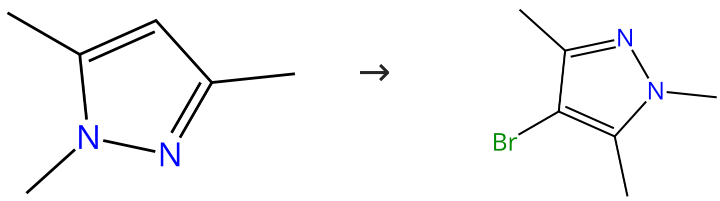
Scheme 36 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (103) Suppliers (61)

31-084-CAS-10038012	Steps: **1** Yield: **99%**	**Halogenation of aromatic compounds by N- chloro-, N-bromo-, and N-iodosuccinimide**
1.1 **Reagents:** *N*-Iodosuccinimide **Catalysts:** Iron chloride ($FeCl_3$) **Solvents:** Acetonitrile; 0.5 h, 25 °C		By: Tanemura, Kiyoshi; et al Chemistry Letters (2003), 32(10), 932-933.

Scheme 37 (1 Reaction) Steps: **1** Yield: **99%**

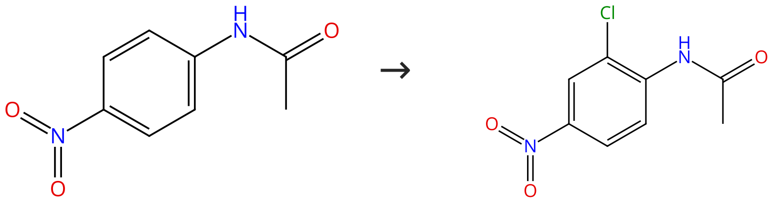


Suppliers (83) Suppliers (82)

31-084-CAS-13835539	Steps: **1** Yield: **99%**	**Halogenation of pyrazoles using N-halosuccinimides in CCl_4 and in water**
1.1 **Reagents:** *N*-Bromosuccinimide **Solvents:** Carbon tetrachloride; 2 h, 25 °C		By: Zhao, Zhi-Gang; et al Synthetic Communications (2007), 37(1), 137-147.
Experimental Protocols		

Scheme 38 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (75)

Suppliers (20)

31-084-CAS-23028293

Steps: **1** Yield: **99%**

1.1 **Reagents:** *N*-Chloro-*N*-(phenylsulfonyl)benzenesulfonamide
Solvents: Acetonitrile; 10 min, 20 - 25 °C

1.2 **Reagents:** Water
Solvents: Dichloromethane; 10 - 15 min, 20 - 25 °C

Experimental Protocols

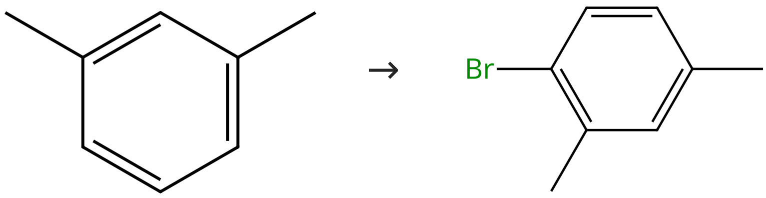
Activator free, expeditious and eco-friendly chlorination of activated arenes by N-chloro-N-(phenylsulfonyl)benzene sulfonamide (NCBSI)

By: Misal, Balu; et al

Tetrahedron Letters (2021), 63, 152689.

Scheme 39 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (97)

Suppliers (81)

31-084-CAS-10073755

Steps: **1** Yield: **99%**

1.1 **Reagents:** *N*-Bromosuccinimide
Catalysts: Gold trichloride
Solvents: 1,2-Dichloroethane; 5 h, 60 °C

Experimental Protocols

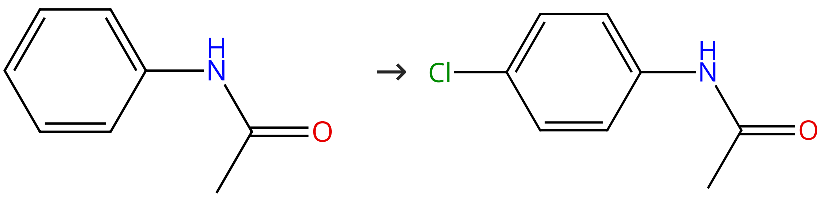
Gold-catalyzed halogenation of aromatics by N-halosucc inimides

By: Mo, Fanyang; et al

Angewandte Chemie, International Edition (2010), 49(11), 2028-2032, S2028/1-S2028/10.

Scheme 40 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (99)

Suppliers (93)

31-084-CAS-23029114

Steps: **1** Yield: **99%**

1.1 **Reagents:** *N*-Chloro-*N*-(phenylsulfonyl)benzenesulfonamide
Solvents: Acetonitrile; 10 min, 20 - 25 °C

1.2 **Reagents:** Water
Solvents: Dichloromethane; 10 - 15 min, 20 - 25 °C

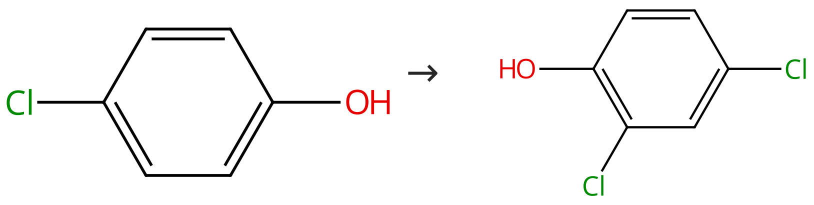
Experimental Protocols

Activator free, expeditious and eco-friendly chlorination of activated arenes by N-chloro-N-(phenylsulfonyl)benzene sulfonamide (NCBSI)

By: Misal, Balu; et al

Tetrahedron Letters (2021), 63, 152689.

Scheme 41 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (90) Suppliers (99)

31-084-CAS-23030492 Steps: **1** Yield: **99%**

1.1 **Reagents:** *N*-Chloro-*N*-(phenylsulfonyl)benzenesulfonamide
Solvents: Acetonitrile; 15 min, 20 - 25 °C

1.2 **Reagents:** Water
Solvents: Dichloromethane; 10 - 15 min, 20 - 25 °C

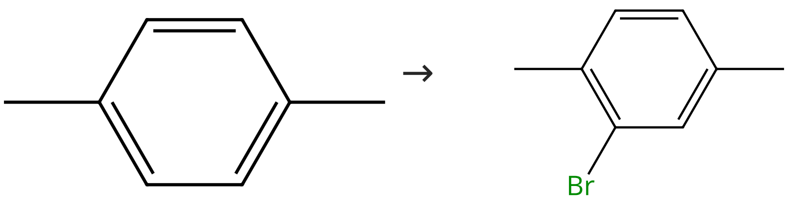
Experimental Protocols

Activator free, expeditious and eco-friendly chlorination of activated arenes by N-chloro-N-(phenylsulfonyl)benzene sulfonamide (NCBSI)

By: Misal, Balu; et al

Tetrahedron Letters (2021), 63, 152689.

Scheme 42 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (115) Suppliers (74)

31-084-CAS-667497 Steps: **1** Yield: **99%**

1.1 **Reagents:** *N*-Bromosuccinimide
Catalysts: Gold trichloride
Solvents: 1,2-Dichloroethane; 30 h, rt

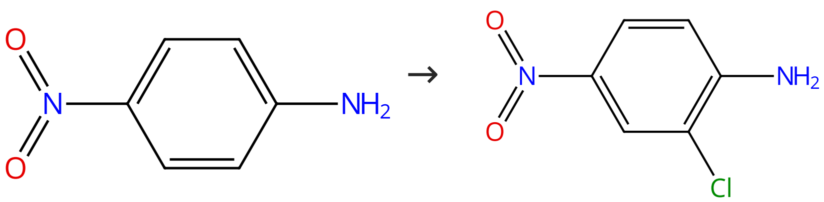
Experimental Protocols

Gold-catalyzed halogenation of aromatics by N-halosucc inimides

By: Mo, Fanyang; et al

Angewandte Chemie, International Edition (2010), 49(11), 2028-2032, S2028/1-S2028/10.

Scheme 43 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (93) Suppliers (63)

31-084-CAS-23027431 Steps: **1** Yield: **99%**

1.1 **Reagents:** *N*-Chloro-*N*-(phenylsulfonyl)benzenesulfonamide
Solvents: Acetonitrile; 5 min, 20 - 25 °C

1.2 **Reagents:** Water
Solvents: Dichloromethane; 10 - 15 min, 20 - 25 °C

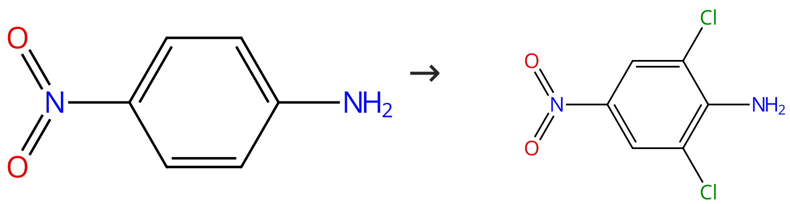
Experimental Protocols

Activator free, expeditious and eco-friendly chlorination of activated arenes by N-chloro-N-(phenylsulfonyl)benzene sulfonamide (NCBSI)

By: Misal, Balu; et al

Tetrahedron Letters (2021), 63, 152689.

Scheme 44 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (93) Suppliers (84)

31-084-CAS-23033407 Steps: **1** Yield: **99%**

1.1 **Reagents:** *N*-Chloro-*N*-(phenylsulfonyl)benzenesulfonamide
Solvents: Acetonitrile; 8 min, 20 - 25 °C

1.2 **Reagents:** Water
Solvents: Dichloromethane; 10 - 15 min, 20 - 25 °C

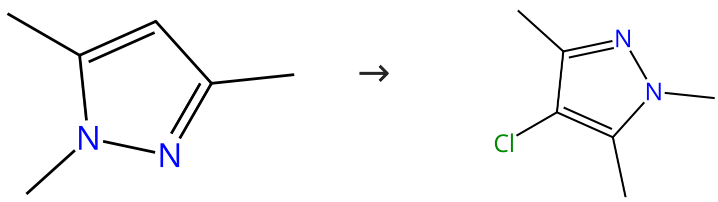
Experimental Protocols

Activator free, expeditious and eco-friendly chlorination of activated arenes by N-chloro-N-(phenylsulfonyl)benzene sulfonamide (NCBSI)

By: Misal, Balu; et al

Tetrahedron Letters (2021), 63, 152689.

Scheme 45 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (83) Suppliers (37)

31-084-CAS-2567980 Steps: **1** Yield: **99%**

1.1 **Reagents:** Chlorosuccinimide
Solvents: Carbon tetrachloride; 1 h, 50 °C

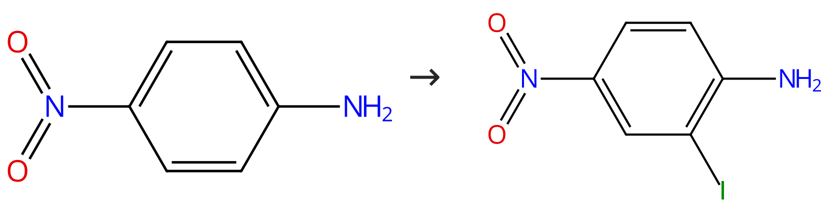
Experimental Protocols

Halogenation of pyrazoles using N-halosuccinimides in CCl_4 and in water

By: Zhao, Zhi-Gang; et al

Synthetic Communications (2007), 37(1), 137-147.

Scheme 46 (1 Reaction) Steps: **1** Yield: **99%**



Suppliers (93) Suppliers (82)

31-084-CAS-9077272 Steps: **1** Yield: **99%**

1.1 **Reagents:** *N*-Iodosuccinimide
Catalysts: [1,1,1-Trifluoro-*N*-[(trifluoromethyl)sulfonyl-κ*O*]methanesulfonamidato-κ*O*]silver
Solvents: Dichloromethane; 1.5 h, 40 °C

Experimental Protocols

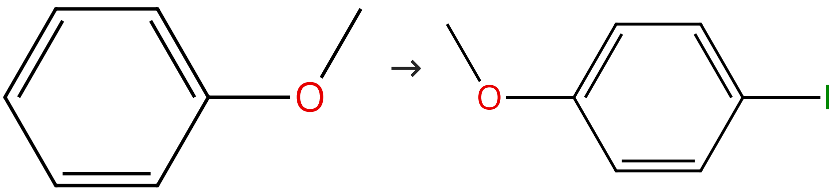
Silver(I)-Catalyzed Iodination of Arenes: Tuning the Lewis Acidity of N-Iodosuccinimide Activation

By: Racys, Daugirdas T.; et al

Journal of Organic Chemistry (2016), 81(3), 772-780.

Scheme 47 (1 Reaction)

Steps: **1** Yield: **99%**



Suppliers (85)

Suppliers (88)

31-084-CAS-9754035

Steps: **1** Yield: **99%**

1.1 **Reagents:** *N*-Iodosuccinimide
Catalysts: Iron chloride ($FeCl_3$)
Solvents: Acetonitrile; 0.5 h, 25 °C

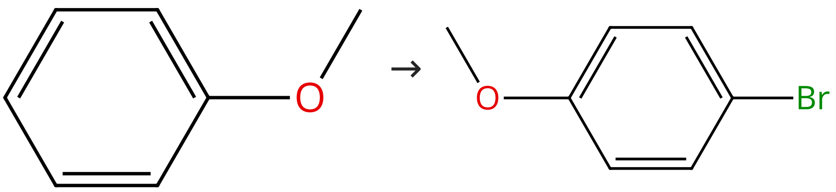
Halogenation of aromatic compounds by N-chloro-, N-bromo-, and N-iodosuccinimide

By: Tanemura, Kiyoshi; et al

Chemistry Letters (2003), 32(10), 932-933.

Scheme 48 (3 Reactions)

Steps: **1** Yield: **99%**



Suppliers (85)

Suppliers (71)

31-614-CAS-44264822

Steps: **1** Yield: **99%**

1.1 **Catalysts:** Iron chloride ($FeCl_3$), 1-Butyl-3-methylimidazolium bis(trifluoromethylsulfonyl)imide; 30 min, rt

1.2 **Reagents:** *N*-Bromosuccinimide
Catalysts: Iodine
Solvents: Acetonitrile; 0.5 h, rt

Experimental Protocols

Dual activation with $Fe(NTf_2)_3$ and I_2: Regioselective halogenation of aromatic compounds using N-halosuccinimides (NXS)

By: Tanemura, Kiyoshi

Tetrahedron (2025), 176, 134544.

31-614-CAS-33508000

Steps: **1** Yield: **99%**

1.1 **Reagents:** *N*-Bromosuccinimide
Catalysts: 3*H*-2,1-Benzoxaselenolium, 3,3-dimethyl-1-phenyl-, tetrafluoroborate(1-) (1:1)
Solvents: Dichloromethane; 23 °C

Experimental Protocols

Hypervalent Chalcogenonium...π Bonding Catalysis

By: Zhang, Qingyu; et al

Angewandte Chemie, International Edition (2022), 61(35), e202208009.

31-084-CAS-3061803

Steps: **1** Yield: **99%**

1.1 **Reagents:** *N*-Bromosuccinimide
Catalysts: Gold trichloride
Solvents: 1,2-Dichloroethane; 1 h, rt

Experimental Protocols

Gold-catalyzed halogenation of aromatics by N-halosuccinimides

By: Mo, Fanyang; et al

Angewandte Chemie, International Edition (2010), 49(11), 2028-2032, S2028/1-S2028/10.